EMC ASSESSMENT

Product: SidecarTridge ROM Emulator

Model/Revision: SidecarTridge ROM Emulator Core Module v3 (Rev.3.0.0) / REV3.0.0

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1. PRODUCT OVERVIEW

The product is a low-voltage digital electronic device intended for use in retro computing systems. It is designed as a core sub-assembly module (connected to a carrier board through castellated edges, mounting holes or pin headers) and is powered at +5 V DC either from its USB-C connector or from an external supply rail provided by the carrier board.

Typical operation voltage is below 50 V DC (e.g. 3.3V / 5V / 12V).

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2. APPLICABLE DIRECTIVE

This assessment is made in relation to Directive 2014/30/EU (Electromagnetic Compatibility - EMC).

The product does not fall under Directive 2014/35/EU (Low Voltage Directive) as it operates below 50 V DC.

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3. DESIGN CHARACTERISTICS RELEVANT TO EMC

- The device contains standard digital logic components (e.g. CMOS/TTL, microcontrollers, memory devices).
- No radio transmitter or intentional RF radiator is present.
- The device does not include wireless communication functionality.
- The design does not incorporate high-frequency switching power supplies (unless provided externally by the host system).
- Signal frequencies are limited to those typical of legacy computing platforms (generally low MHz range or below).
- External connections are limited in length and are typically internal to the host system or via short cables.
- The device includes a microcontroller-based system operating at moderate clock frequencies typical of embedded systems.
- A USB interface is present for local data transfer; it is not intended for continuous high-speed communication during normal operation.

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4. EMISSION ASSESSMENT

Due to the design characteristics described above:

- The product is not expected to generate significant electromagnetic emissions.

- Any emissions are comparable to those of standard digital logic circuits used in legacy computing systems.
- The absence of high-frequency switching regulators and RF transmitters significantly reduces emission risk.
- The USB interface is used intermittently for file transfer and does not represent a continuous high-frequency emission source.

Conclusion:
The product presents a low risk of causing electromagnetic interference when used as intended.

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5. IMMUNITY ASSESSMENT

- The product is designed to operate within the electromagnetic environment provided by the host system.
- Immunity performance depends on the overall system in which the product is installed.
- The product does not include specific shielding or filtering beyond standard design practices.

Conclusion:
The product is expected to operate correctly in typical residential, commercial, and light-industrial environments when integrated into a compliant host system.

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6. INSTALLATION AND USE CONDITIONS

- The product is intended for integration into a larger electronic system as a carrier-board sub-assembly.
- It is not designed to operate as a standalone end-user device; overall conformity of the finished apparatus is the responsibility of the assembler placing it on the market.
- Proper operation and EMC performance depend on correct installation in a compliant carrier-board / host-system design.

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7. OVERALL CONCLUSION

Based on the design characteristics and intended use, the product is considered to comply with the essential requirements of Directive 2014/30/EU (EMC), provided that it is installed and used in accordance with its intended purpose.

This assessment is supported by the use of harmonised standards such as:

- EN IEC 61000-6-1:2019 (Immunity)
- EN IEC 61000-6-3:2021 (Emissions)

No additional EMC mitigation measures are required based on the design and intended use.

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8. RESPONSIBLE PERSON

Name: Diego Parrilla Santamaria
Position: CEO
Company: GOODDATA LABS SL

Date: 23 April 2026

Signature: